

ENSC 225 – Lab #4: Characterization of MOSFETS and MOS Amplifier

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Abstract

This laboratory experiment focuses on the characterization and analysis of MOSFETs and a MOS amplifier. The key objective is to determine the threshold voltage (V_t) and transconductance (K) for both n-type and p-type MOSFETs, utilizing Static Power Analyzer (SPA) scans. The lab provides a detailed procedure for extracting these parameters from I_D vs V_{GS} and I_D vs V_{DS} characteristics, including the analysis of early voltage and drain output resistance (r_o) through experimental plots. The MOSFETs are also tested for electrostatic sensitivity, with precautionary measures to avoid damage from static discharge.

In addition to MOSFET characterization, the lab involves building and analyzing a simple common-source (CS) NMOS amplifier, determining its DC operating points, and experimentally computing its voltage gain. The amplifier's performance is further assessed by introducing a small signal and evaluating its frequency response from 10Hz to 2MHz. The impact of a source bypass capacitor (CS) on the amplifier's behavior is also studied. Theoretical calculations for gain, input impedance, and output impedance are compared with experimental results, and the frequency response is used to identify the amplifier's cut-off frequencies. The lab emphasizes data analysis, model comparison, and the importance of clear, structured reporting of experimental results.

I_D versus V_{GS} Characteristics for the determination of V_t and K

Introduction: In this part of the experiment, we determine the threshold voltage (V_t) and transconductance (K) for both n-type (n-MOS) and p-type (p-MOS) MOSFETs by analyzing their I_D vs. V_{GS} characteristics. By plotting the square root of I_D versus V_{GS} for different substrate voltages, we can extract V_t and calculate the transconductance parameters (K_n for n-MOS and K_p for p-MOS). This allows us to understand how substrate bias influences the MOSFET's performance.

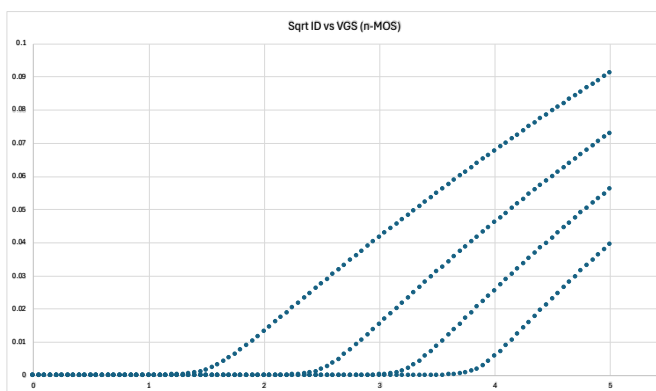


Figure 1: Sqrt ID vs VGS (n-MOS)

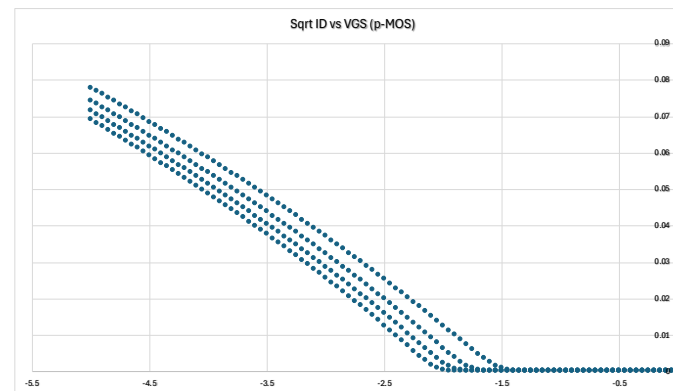


Figure 2: Sqrt ID vs VGS (p-MOS)

n-MOS Calculations

Substrate Voltage (V)	V_T (V)	$K_n(\frac{mA}{V^2})$
0	1.1	1.096
-1	2.15	1.307
-2	3.0	1.550
-3	3.5	1.191

p-MOS Calculations

Substrate Voltage (V)	V_T (V)	$K_p(\frac{mA}{V^2})$
0	-1.2	0.859
1	-1.5	0.928
2	-1.7	0.970
3	-1.9	1.033

Discussion: The graphs show the relationship between drain current (I_D) and gate-source voltage (V_{GS}) for n-MOS and p-MOS transistors under different substrate bias conditions. These plots allow extracting the key transistor parameters of threshold voltage (V_T) and transconductance factor (K_n or K_p). For n-MOS, V_T increases as substrate voltage (V_{SB}) becomes more negative, due to the body effect. However, K_n remains constant. Conversely, for p-MOS, V_T decreases with more negative V_{SB} , as the body effect lowers effective channel doping, while K_p increases. Understanding these substrate bias effects is crucial for modeling and designing MOS circuits, as summarized in the data table.

I_D versus V_{DS} Characteristics for the determination of r_o

Introduction: In this section, we focus on plotting the I_D versus V_{DS} characteristics for both n-type (n-MOS) and p-type (p-MOS) MOSFETs to determine key parameters such as the early voltage and the drain output resistance (r_o). By analyzing the response curve that shows the maximum drain current (I_D), we can extract the early voltage and calculate r_o at the operating point. These parameters are essential for understanding the MOSFET’s behavior in amplification, as they influence the gain and performance of the amplifier in the following section of the experiment.

Plots

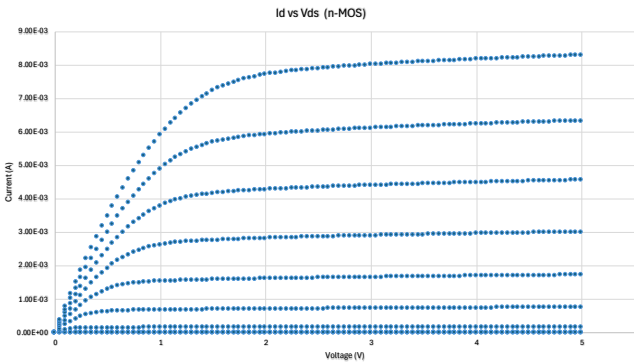


Figure 3: I_D vs V_{DS} (n – MOS)

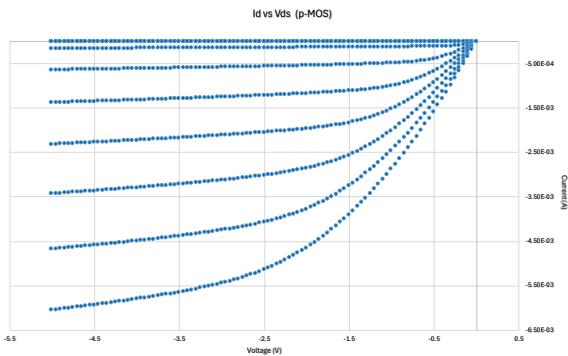


Figure 4: I_D vs V_{DS} (p – MOS)

Calculations for the drain output resistance (r_o) and early voltage (V_A):

n-MOS

$$r_o = \frac{1}{\text{slope}}$$

Slope Coordinates: (4.45, 0.00823) and (2.5, 0.00788)

$$\text{Slope} = \frac{0.00823 - 0.00788}{4.45 - 2.45} = 0.175 \times 10^{-3}$$

$$r_o = \frac{1}{0.175 \times 10^{-3}} = 5714\Omega$$

$$r_o = \frac{V_A + V_{DS}}{I_D}$$

Slope Coordinate: (4.45, 0.00823)

$$V_A = r_o I_D - V_{DS}$$

$$V_A = (5714)(0.00823) - (4.45)$$

$$V_A = 42.5433V$$

	Drain Output Resistance (r_o)	Early Voltage (V_A)
n-MOS	5714 Ω	42.5433V
p-MOS	3148 Ω	-14.125V

This section focuses on determining the drain output resistance (r_o) and early voltage (V_A) for both n-MOS and p-MOS transistors. The graphs show the I_D versus V_{DS} characteristics, which are used to extract these important parameters. The equations provided demonstrate how the slope and x-intercept of the linear region of these curves can be used to calculate r_o and V_A , respectively. The data table summarizes the calculated values for n-MOS and p-MOS devices. Understanding drain output resistance and early voltage is crucial, as they impact the output resistance and gain of MOS-based circuits. The changes in these parameters with substrate bias, as seen in the previous section, illustrate how circuit performance can be influenced by the selection of operating voltages. This comprehensive transistor characterization enables better modeling and design of MOS integrated circuits.

Experimental determination of V_t and K

Introduction: In this part of the experiment, we will determine the threshold voltage (V_t) and transconductance (K) for both n-type (NMOS) and p-type (PMOS) MOSFETs using a simple set of voltage measurements and calculations. By setting up a circuit with a 5V single supply and a 1k Ω resistor, we will measure the voltage across different points in the circuit and use these readings to calculate V_t and K. The procedure involves measuring the supply voltage, the voltage at specific nodes, and the voltage changes with the addition of the resistor in parallel to the Digital Multimeter (DMM). These measurements will allow us to experimentally determine the threshold voltage and transconductance values for both NMOS and PMOS devices, which will then be compared with results obtained in previous sections.

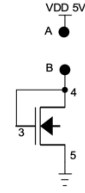


Figure1

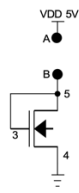


Figure2

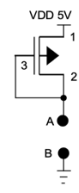


Figure3

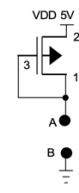


Figure4

	Supply Voltage V_{DD}	Node V_{AB}	Node V_{AB} w/Resistor	Resistor value
Figure 1	5.0001V	4.371V	1.5677V	0.9845k Ω
Figure 2	5.000V	4.352V	1.5661V	0.9845k Ω
Figure 3	5.000V	3.568V	1.4185V	0.9845k Ω
Figure 4	5.000V	3.570V	1.4399V	0.9845k Ω

n-MOS V_t and K Calculations:

Figure 1: $i_D = \frac{1}{2} K_n (V_t)^2$ $\frac{1.5677V}{0.9845kW} = \frac{1}{2} K_n (V_t)^2$	Figure 2: $i_D = \frac{1}{2} K_n (V_{AB} - V_t)^2$ $\frac{1.5661V}{0.9845kW} = \frac{1}{2} K_n (4.352 - V_t)^2$
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$$V_t = 2.176, K_n = 0.672 \times 10^{-3}$$

p-MOS V_t and K Calculations:

Figure 3: $i_D = \frac{1}{2} K_n (V_t)^2$ $\frac{1.4185V}{0.9845kW} = \frac{1}{2} K_n (V_t)^2$	Figure 4: $i_D = \frac{1}{2} K_n (V_{DD} - V_{AB} - V_t)^2$ $\frac{1.4399V}{0.9845kW} = \frac{1}{2} K_n (5.000 - 3.570 - V_t)^2$
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$$V_t = 0.715, K_n = 5.477 \times 10^{-3}$$

Discussion: This part of the experiment determines the threshold voltages (V_t) and transconductance factors (K_n , K_p) of n-MOS and p-MOS devices. It specifies the supply voltage, node voltages, and resistor values used in the circuit to make these measurements. The subsequent equations and parameters derived demonstrate the methodologies for calculating V_t and K for both n-MOS and p-MOS transistors. Comparing these calculated values in the data tables to the graphical plots shown in section 1 allows us to understand how substrate bias impacts the threshold and transconductance characteristics of the n-MOS and p-MOS devices. This comprehensive characterization is essential for properly modeling and designing MOS-based circuits that will operate reliably under different substrate voltage conditions.

Resistive Load, common source NMOS amplifier

Introduction: In this part of the lab, we will construct a common-source NMOS amplifier with a resistive load, using $10\text{k}\Omega$ for the drain resistor (R_D) and $10\text{M}\Omega$ for the gate resistor (R_G). After measuring key voltages and calculating the DC current (I_D), we will apply a small 1kHz sinusoidal input to the amplifier and observe the output to compute the gain. We will also investigate the effect of increasing input signal amplitude on distortion, measure the peak-to-peak output, and capture the waveform. Finally, we will perform a frequency response sweep from 10Hz to 2MHz and compare the experimental gain with the theoretical gain based on the small-signal model.

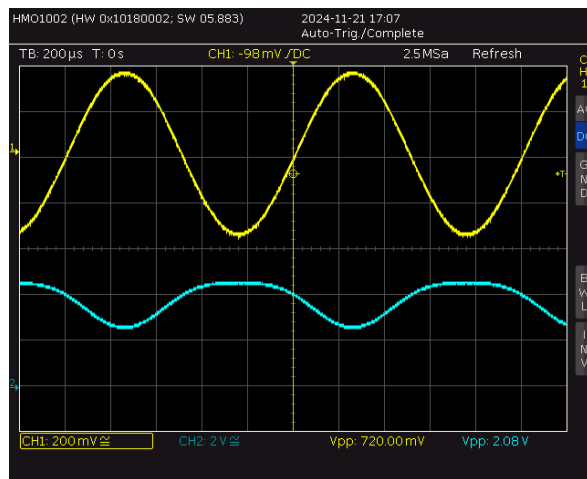
R_D	R_G
$9.884\text{k}\Omega$	$9.854\text{M}\Omega$

Measured V_{DD}	Measured V_D
5.000V	2.8828V

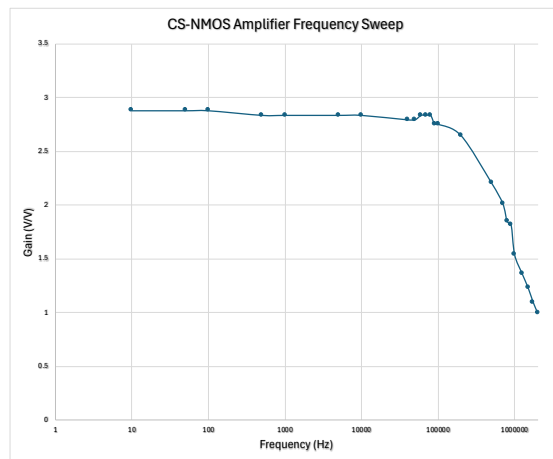
Computing DC I_D :

$$I_D = \frac{(V_{DD} - V_D)}{R_D} = \frac{5.000\text{V} - 2.8828\text{V}}{9.884\text{k}\Omega} = 0.214\text{mA}$$

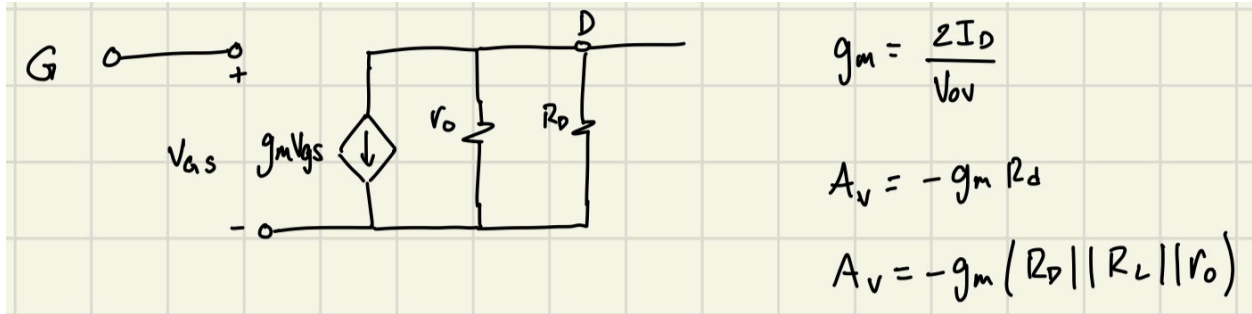
Input Voltage	Output Voltage	Gain
56mV	160mV	$\frac{160\text{mV}}{56\text{mV}} = 2.857$



Distortion shows at 0.33V and Pk-Pk output is 2.08V



CS-MOS Amplifier Frequency Sweep



	Theoretical	Measured
Gain(A_v)	2.91	2.84

Discussion: This section of the lab focuses on constructing and analyzing a common-source NMOS amplifier circuit. The key aspects include measuring the DC input and output voltages, as well as the gain of the amplifier. The provided equations are then used to calculate the gain and transconductance of the amplifier. Additionally, the frequency response of the amplifier is observed through the AC waveform plot. This comprehensive characterization, which also involves comparing the measured and theoretical values for the amplifier parameters, is crucial for understanding the behavior and potential applications of the NMOS amplifier in analog circuit design. The lab setup and measurements allow for a thorough evaluation of the amplifier's performance, equipping the students with the necessary knowledge to design and optimize NMOS-based amplifier circuits.

CS Amplifier with Capacitor Coupling

Introduction: In this section we analyzed a CS Amplifier with and without a bypass capacitor, while keeping the transistor in saturation, we then measured and calculated operating points, voltages, gain, input and output impedences for both of the situations.

DMM Measured Component Values

Component	Measured Value
R_D	2.6723 k Ω
R_S	0.9845 k Ω
R_{G1}	98.23 k Ω
R_{G2}	98.59 k Ω
R_L	26.86 k Ω
C_{C2}	8.2577 μ F
C_{C1}	8.8396 μ F
C_S	27.039 μ F

Calculating $V_{DS(off)}$ and $I_{D(sat)}$:

$V_{DS(off)} : V_{GS} \leq I_{tn}, I_D = 0$ $V_G - V_S \leq V_t$ $5.987 - V_S \leq 0.715$ $V_S \leq 5.272$ $V_{DS} = V_D - V_S = 12 - 5.272$ $V_{DS} = 6.728V$	$I_{D(sat)} : V_{GS} \leq I_{tn}, V_{DS} \leq V_{GS} - V_{tn}$ $V_D - V_S \geq V_G - V_S - V_{tn}$ $V_D \geq V_G - V_{tn}$ $I_{D(sat)} R_D \geq V_G - V_{tn}$ $I_{D(sat)} \geq \frac{V_G - V_{tn}}{R_D}$ $= \frac{5.987 - 0.715}{2.6723}$ $I_{D(sat)} = 1.97283mA$
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	procedure	Qpoint					Voltage gain		A_V	R_{in} k Ω	R_{out} k Ω
		V_G	V_D	V_S	$V_{DS}(V)$	$I_D(mA)$	$V_{in}(mV)$	$V_{out}(mV)$			
W/O C_E	theoretical	5.989	6.333	2.046	2.285	2.37	///	///	1.067	50	1.717
	measured	5.987	7.846	1.442	6.397	1.554	2.32	2.24	0.965	48.726	2.255
W/ C_E	theoretical	5.989	6.333	2.046	2.285	2.37	///	///	2.78	49.2	1.706
	measured	5.987	7.749	1.442	6.397	1.554	2.24	6.64	2.84	48.726	2.289

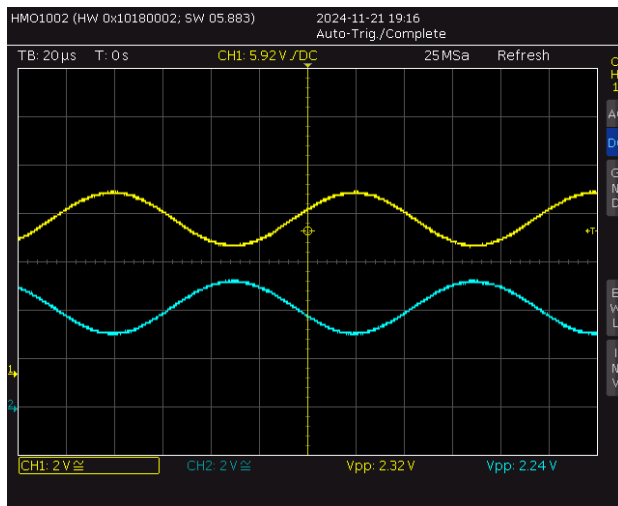


Figure 1: **Cs not connected**

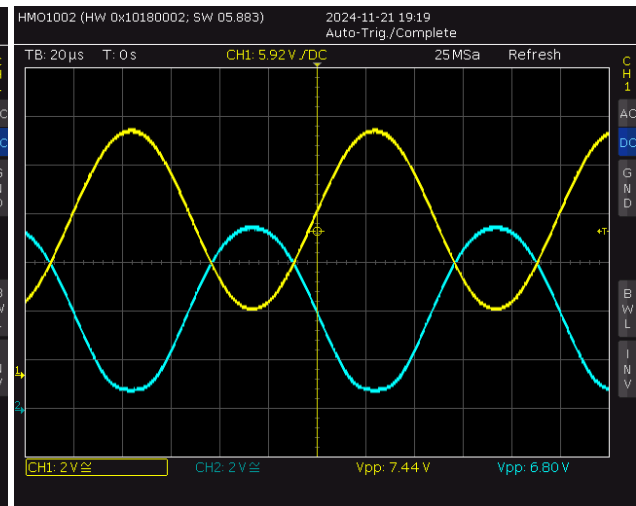


Figure 2: **Output Distortion at 3.36V, Vpp at 6.80V**

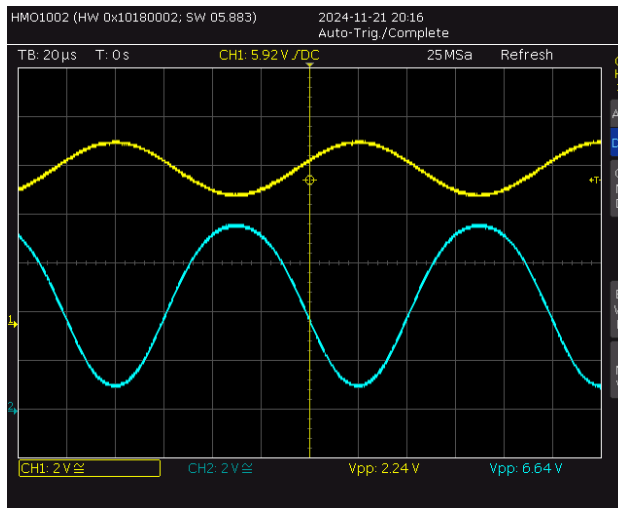


Figure 3: **Cs connected across Rs-Bypass**

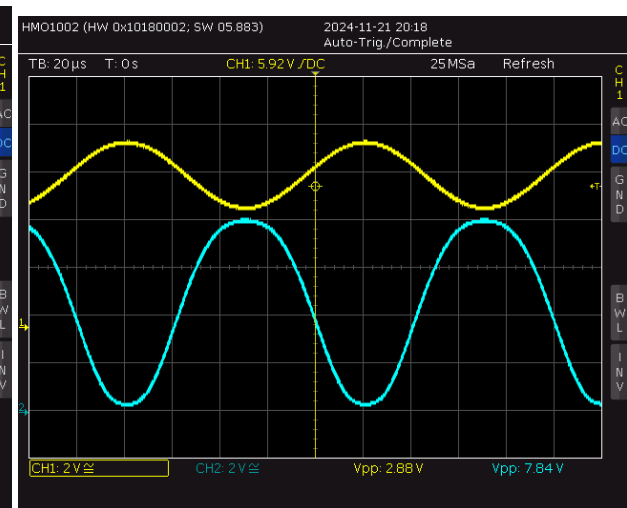
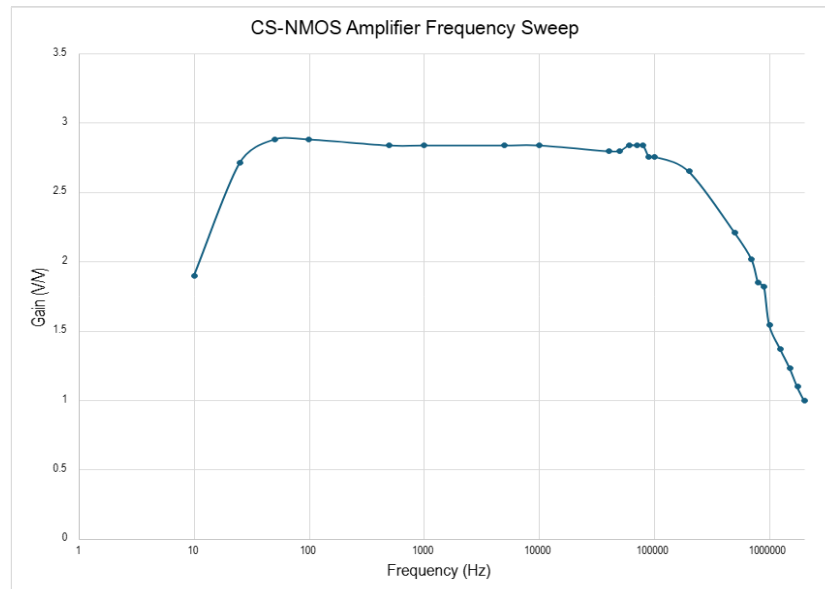


Figure 4: **Output Distortion at 1.26V, Vpp = 7.84V**

Frequency Response Measurement



The frequency response of our amplifier exhibits a bandpass characteristic, with a low-frequency cutoff around 100 Hz caused by the source coupling capacitor, which attenuates low-frequency signals due to its increasing impedance at lower frequencies. The high-frequency cutoff near 100 kHz is due to the FET's internal parasitic capacitances, which act as a low-pass filter, reducing gain at higher frequencies. Together, these factors define the amplifier's frequency range, allowing it to amplify signals within this band while attenuating frequencies outside it.

Final Discussion

Discrepancies between theoretical and practical results in this lab arise from measurement limitations, device imperfections, and environmental effects. Oscilloscopes and DMMs have finite precision, introducing errors in small-signal or high-frequency measurements. Sometimes the oscilloscope wave will be fluctuating with smaller voltages, making it more difficult to extract an exact value. MOSFET parameters like threshold voltage (V_t) and transconductance (K) can vary due to manufacturing tolerances and thermal effects, while parasitic capacitances and stray inductances affect frequency-dependent behavior. Non-ideal resistors and capacitors with tolerances further shift bias points and cutoff frequencies. Signal distortion at higher amplitudes and second-order effects like channel-length modulation also contribute, highlighting the need to consider real-world factors when analyzing circuits.